

SURAKSHA

A N I N T E L L I G E N T V U L N E R A B I L I T Y S C A N N E R

Name : K PRANAV ESWAR

Reg No. : LMC25MCA-2034

Dept : S3 MCA

GUIDE : Ms Bismi K Charleys



TRUST

INTRODUCTION

A WEB-BASED CYBERSECURITY VULNERABILITY ASSESSMENT PLATFORM.

It combines different security checks in a single system:

- Web Vulnerability Scanning
- Network Reconnaissance
- SSL/TLS Security Analysis
- Protocol Analysis
- CVE Mapping
- ML-Based Risk Prediction

Flow in simple words:

Detect → Analyse → Assess → Report

LITERATURE SURVEY

No.	Paper Title	Author	Concept	Limitations
1	AI-Powered Vulnerability Detection and Patch Management in Cybersecurity: A Systematic Review of Techniques, Challenges, and Emerging Trends (2026)	Malkawi & Alhajj	Reviews AI, ML, DL, LLM and DRL techniques for vulnerability detection, prioritization and patch management.	Dataset limitations, lack of standard benchmarks, limited generalizability, poor explainability and lack of adversarial testing.
2	UniEmbed: A Novel Approach to Detect XSS and SQL Injection Attacks Leveraging Multiple Feature Fusion with Machine Learning Techniques (2025)	Bakir	Detects XSS and SQL Injection using Word2Vec, FastText, USE and machine-learning classifiers.	Mainly evaluated using benchmark datasets; real-world generalization needs further validation.
3	Detecting SQL Injection Attacks by Binary Gray Wolf Optimizer and Machine Learning Algorithms (2024)	Arasteh	Uses Binary Gray Wolf Optimizer, Artificial Neural Network and Decision Tree for SQL Injection detection.	The performance of learning-based SQLi detection depends on the quality of the training data and the learning method, which can lead to false-positive and false-negative results.

EXISTING SYSTEM

- Separate tools for different security assessments.
- Web and network vulnerabilities analyzed independently.
- Mostly rule/signature-based detection.
- Limited automated risk prioritization.
- Manual analysis of security reports is often required.

PROBLEM STATEMENT

- Requires multiple specialized tools for cybersecurity assessment.
- Raw technical outputs - difficult for non-experts to understand.
- Vulnerability detection - not indicate risk or priority.
- Security findings - fragmented across different assessment areas.

RESEARCH GAP/STUDY

- Existing research - focuses on individual security assessment techniques - often developed independently.
- An integrated platform combining multiple security analyses with ML-based risk prediction - is needed.

OBJECTIVES

- Develop the integrated cybersecurity platform SURAKSHA.
- Perform web vulnerability scanning and network reconnaissance.
- Analyze SSL/TLS security and network protocols.
- Map vulnerabilities to relevant CVE records.
- Predict security risk levels using machine learning.
- Provide centralized, risk-based security findings.

PROPOSED SYSTEM

SURAKSHA is an integrated cybersecurity assessment platform that combines multiple security analysis techniques with ML-based risk prediction.

System includes :

- Performs web vulnerability scanning for SQL Injection, XSS, security headers and sensitive paths.
- Performs network reconnaissance using Nmap for ports, services and versions.
- Checks SSL/TLS configuration and certificate details.
- Performs protocol and service analysis.
- Maps detected services and vulnerabilities to CVE information.
- Uses Machine Learning to support risk prediction.
- Stores scan results and vulnerabilities in a centralized SQLite database.
- Generates reports with severity, risk, evidence and recommendations.

TECHNOLOGY STACK

Frontend : HTML · CSS · JavaScript · Bootstrap

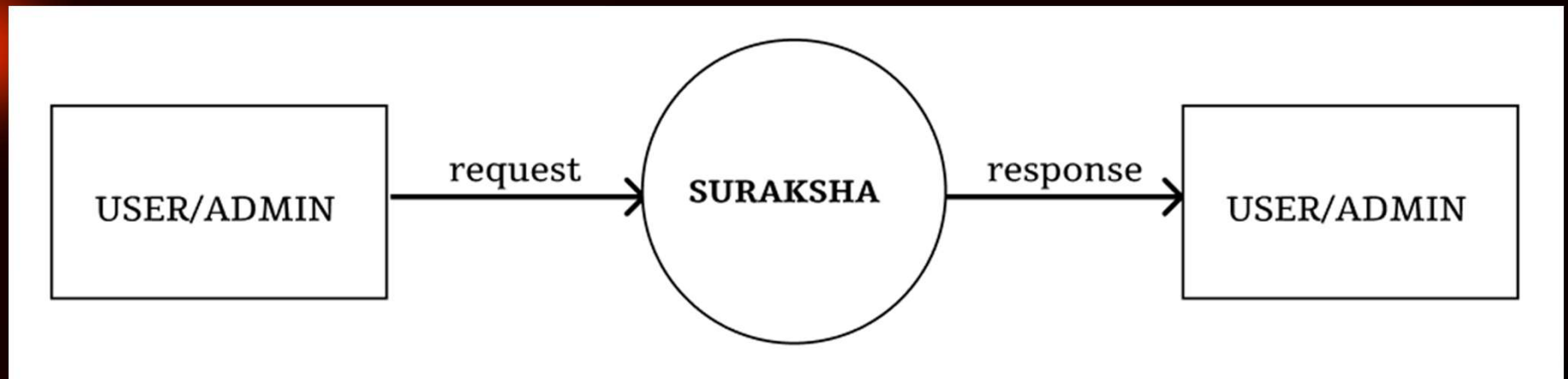
Backend : Python · Flask

Database : SQLite3

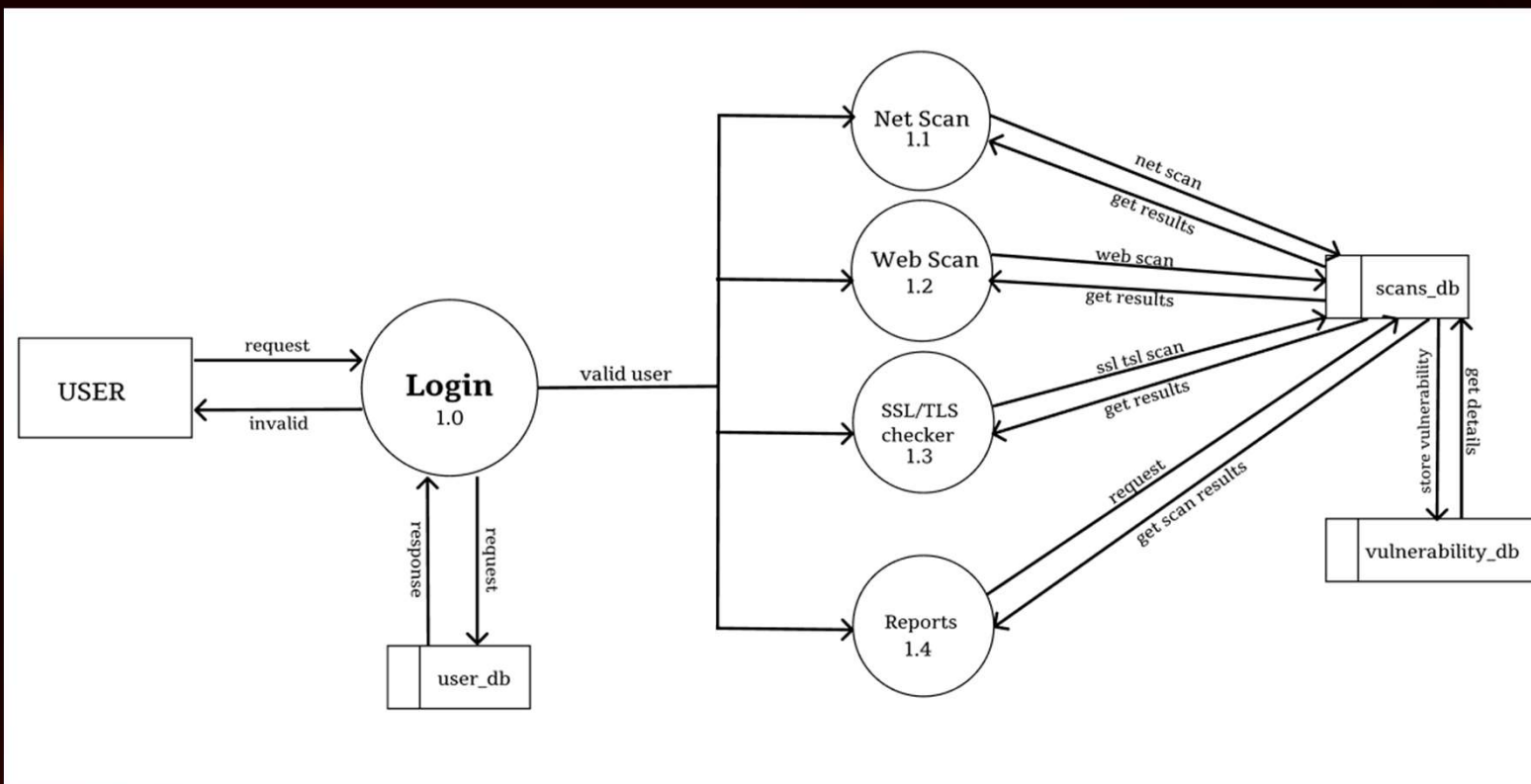
Machine Learning : Scikit-learn · Joblib · Random Forest

Security Tools : Nmap · Python-Nmap · Requests

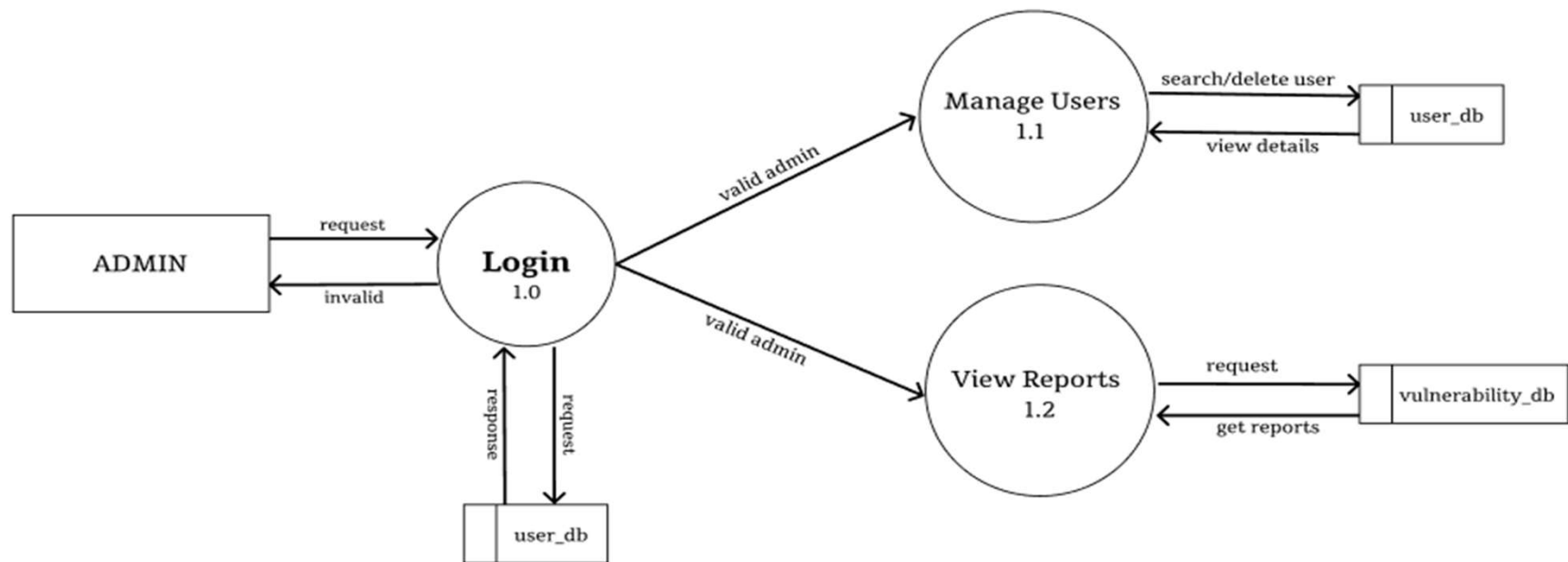
CONTEXT DIAGRAM - LEVEL 0



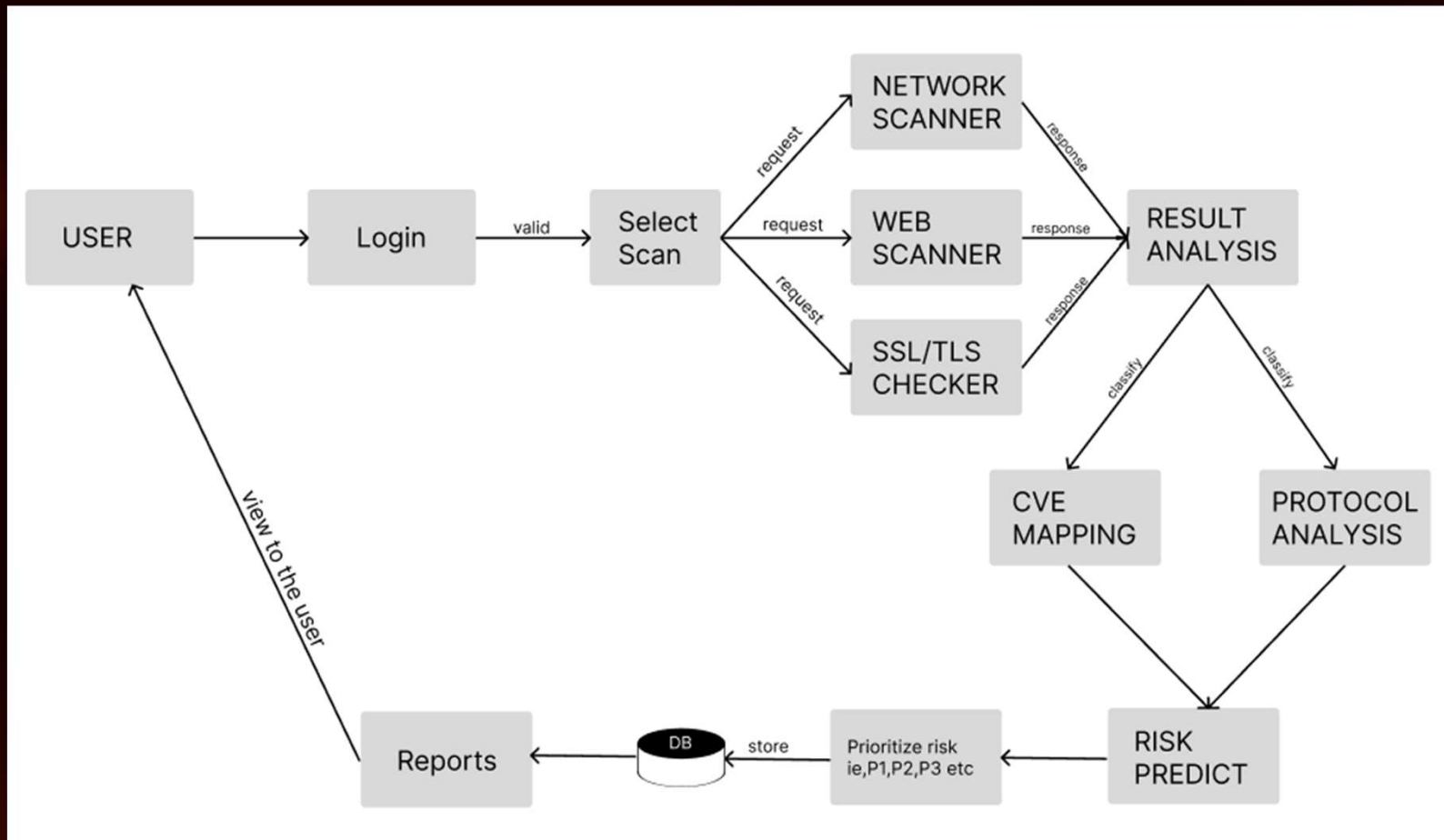
USER - DFD - LEVEL 1



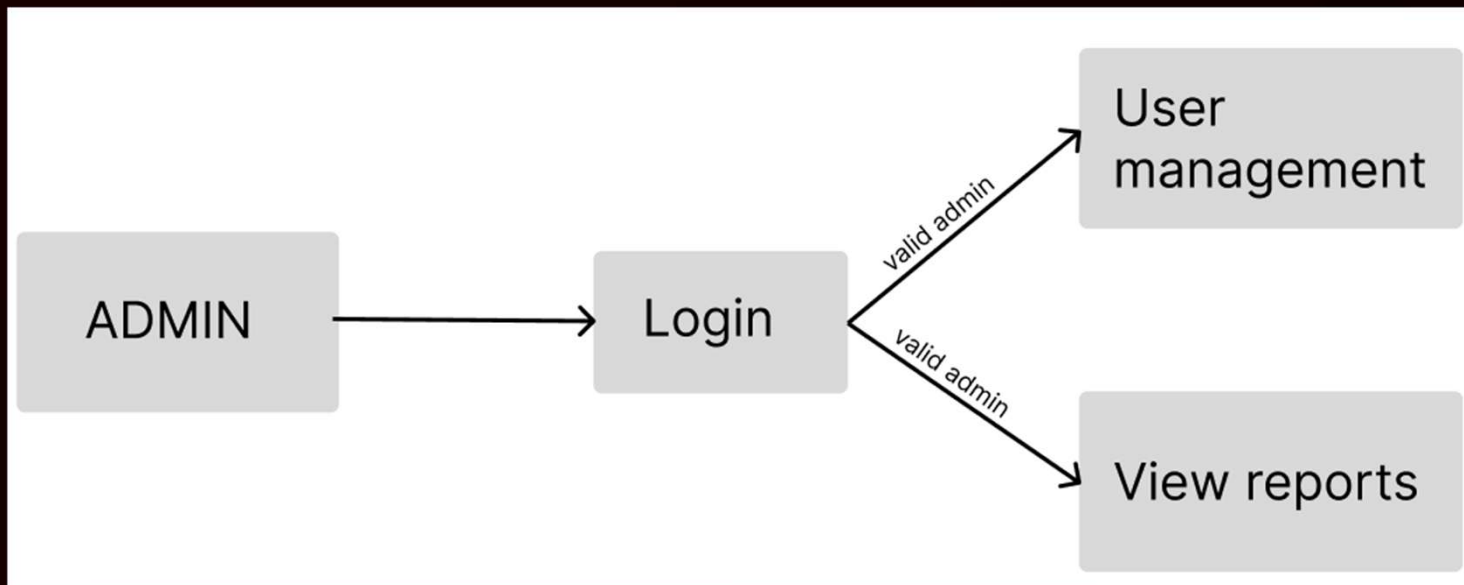
ADMIN - DFD - LEVEL 1



USER - ARCHITECTURAL FLOW



ADMIN - ARCHITECTURAL FLOW



DATABASE DESCRIPTION

Schema : scanner.db

Tables : user,scan,vulnerability

User table

Column Name	Data Type	Constraints	Description
id	INTEGER	PRIMARY KEY, NOT NULL	Unique user ID
username	VARCHAR(100)	NOT NULL	User's name
email	VARCHAR(120)	NOT NULL, UNIQUE	User's email address
password	VARCHAR(255)	NOT NULL	User's password or password hash

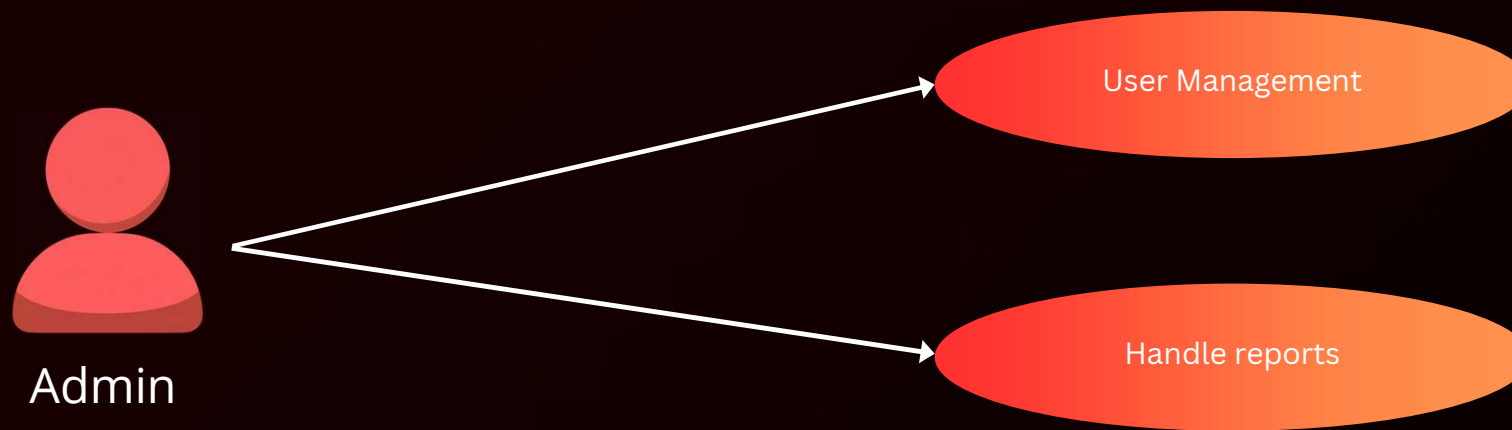
scans table

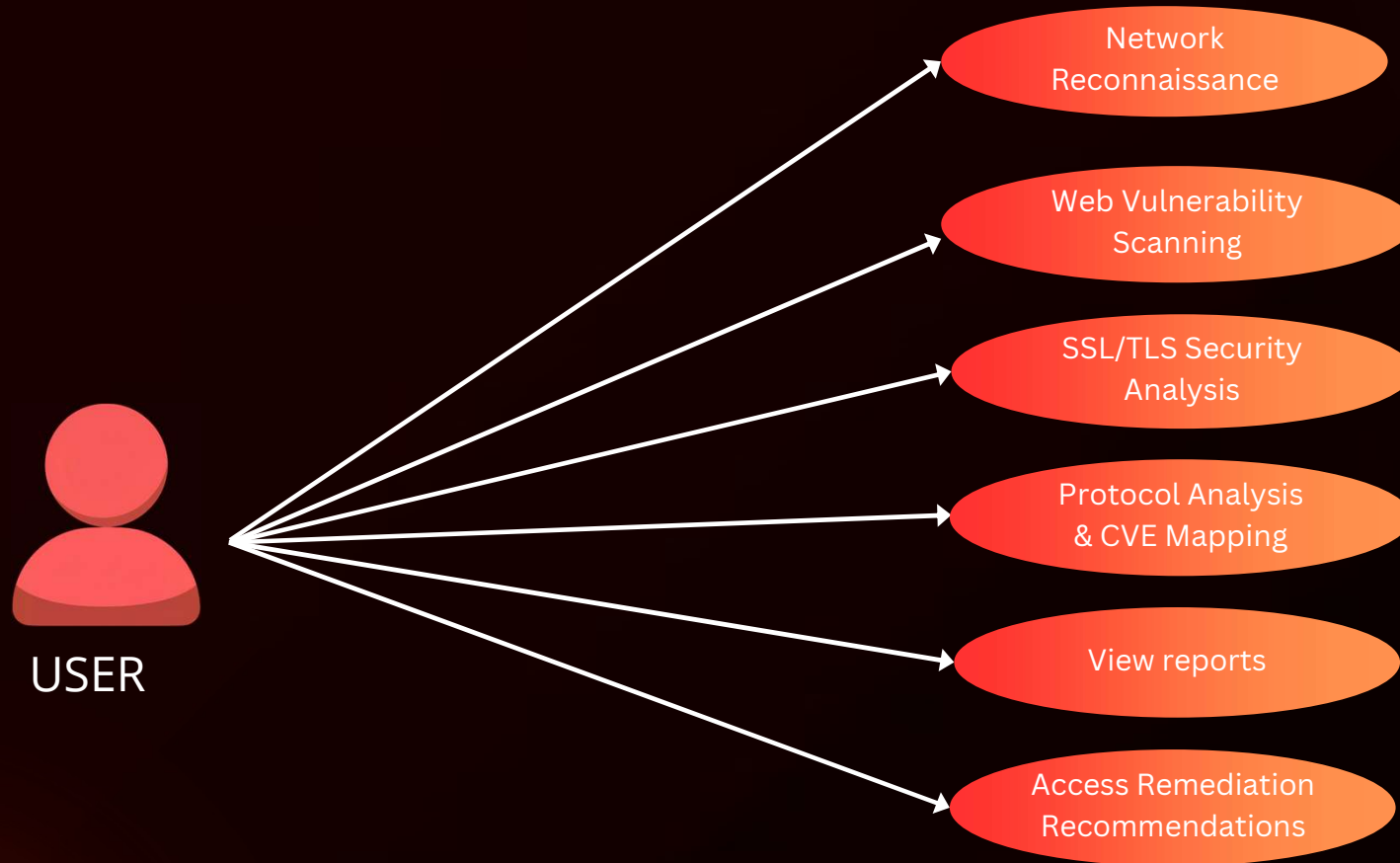
Column Name	Data Type	Constraints	Description
id	INTEGER	PRIMARY KEY, NOT NULL	Unique scan ID
target	VARCHAR(255)	NOT NULL	Scanned target
scan_type	VARCHAR(100)	NOT NULL	Type of scan
severity	VARCHAR(50)	NOT NULL	Overall scan severity
status	VARCHAR(50)	NULLABLE	Scan status
open_ports	INTEGER	NULLABLE	Number of open ports
details	TEXT	NULLABLE	Scan results and additional information
user_id	INTEGER	NOT NULL, FOREIGN KEY	ID of the user who performed the scan
created_at	DATETIME	NULLABLE	Scan creation date and time

vulnerability table

Column Name	Data Type	Constraints	Description
id	INTEGER	PRIMARY KEY, NOT NULL	Unique vulnerability ID
port	INTEGER	NOT NULL	Affected port number
service	VARCHAR(100)	NOT NULL	Network service
protocol	VARCHAR(50)	NULLABLE	Network protocol
vulnerability_name	VARCHAR(255)	NOT NULL	Name of the vulnerability
cve_id	VARCHAR(100)	NULLABLE	CVE identifier
severity	VARCHAR(50)	NOT NULL	Vulnerability severity
attack_type	VARCHAR(255)	NULLABLE	Type of attack
description	TEXT	NULLABLE	Vulnerability description
recommendation	TEXT	NULLABLE	Recommended solution
scan_id	INTEGER	NULLABLE, FOREIGN KEY	Associated scan ID

MODULE DESCRIPTION





RISK PREDICTION MODULE

ALGORITHM & CONFIGURATION

PARAMETER	VALUE
Algorithm	Random Forest
Number of Trees (n_estimators)	200
Maximum Tree Depth (max_depth)	10
Random State	42
Test Size	20%
Number of Input Features	6
Output Classes	4

TRAINING DATASET



low



medium



high



critical

SCREENSHOTS

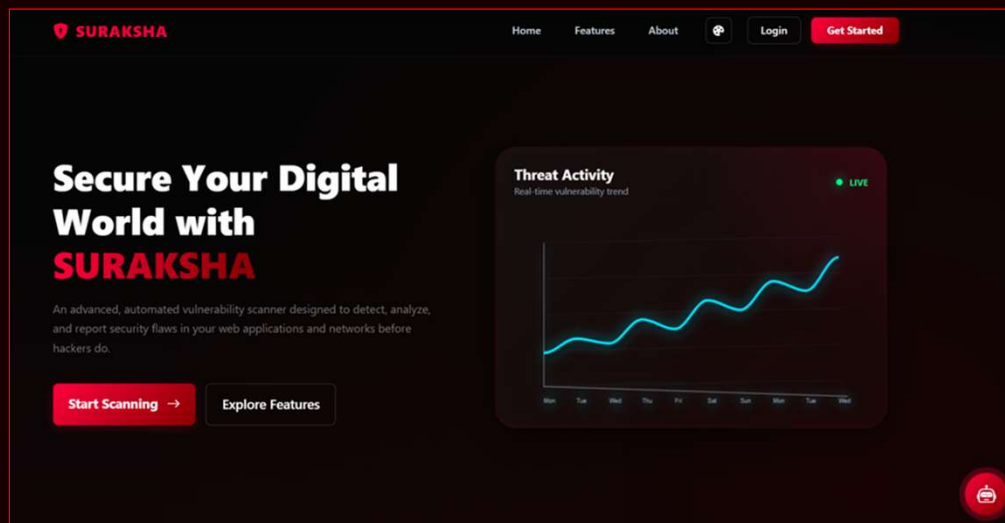


fig. home page

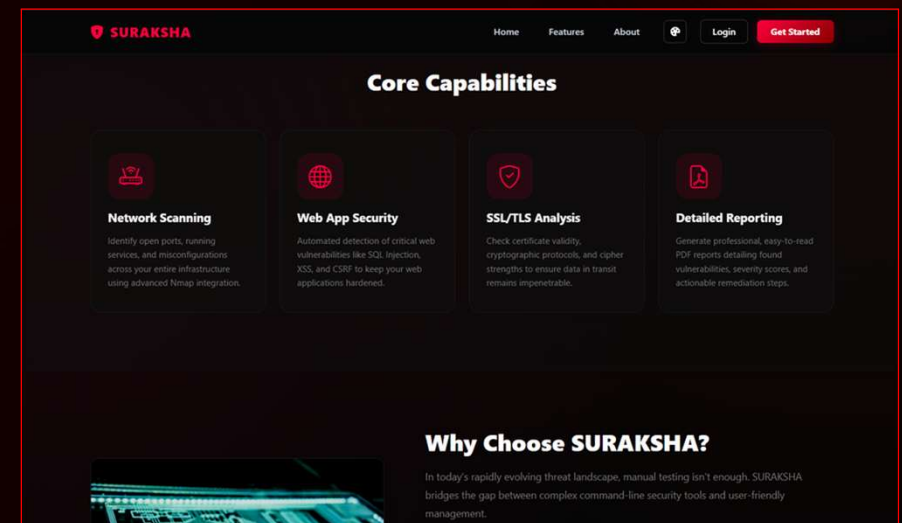


fig. features page

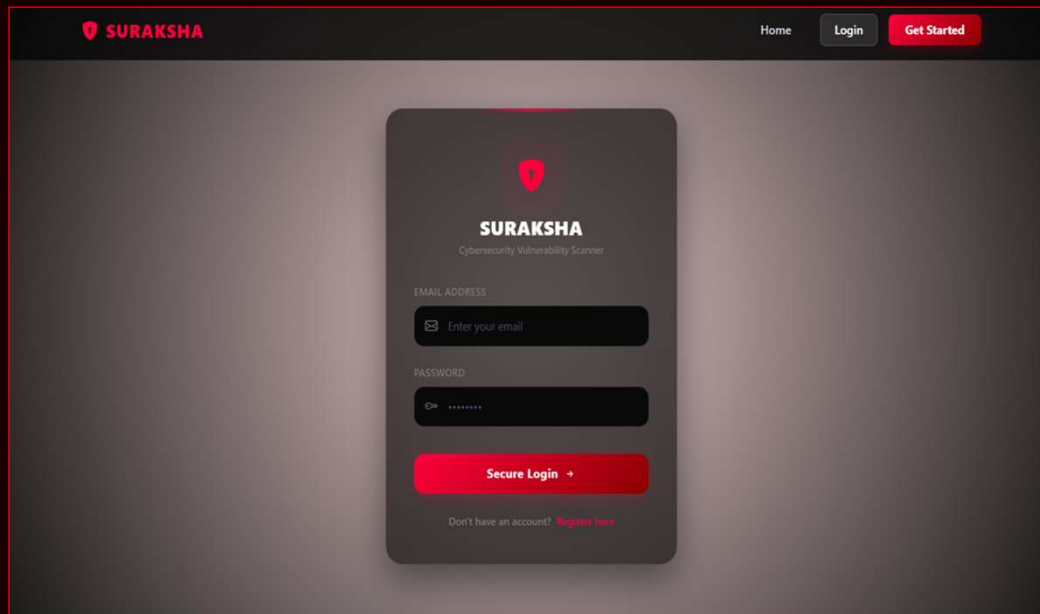


fig. login page

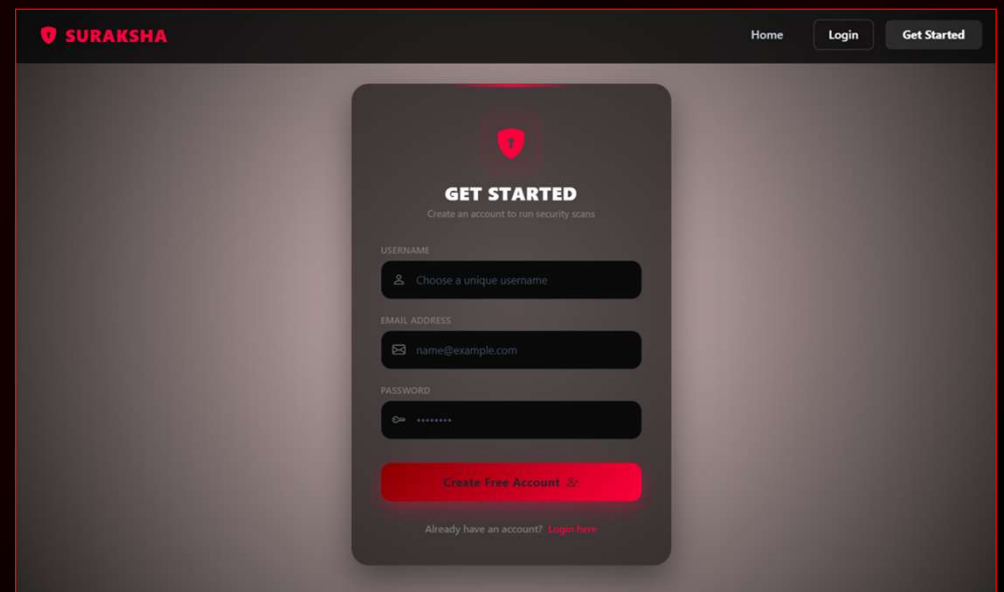


fig. register page

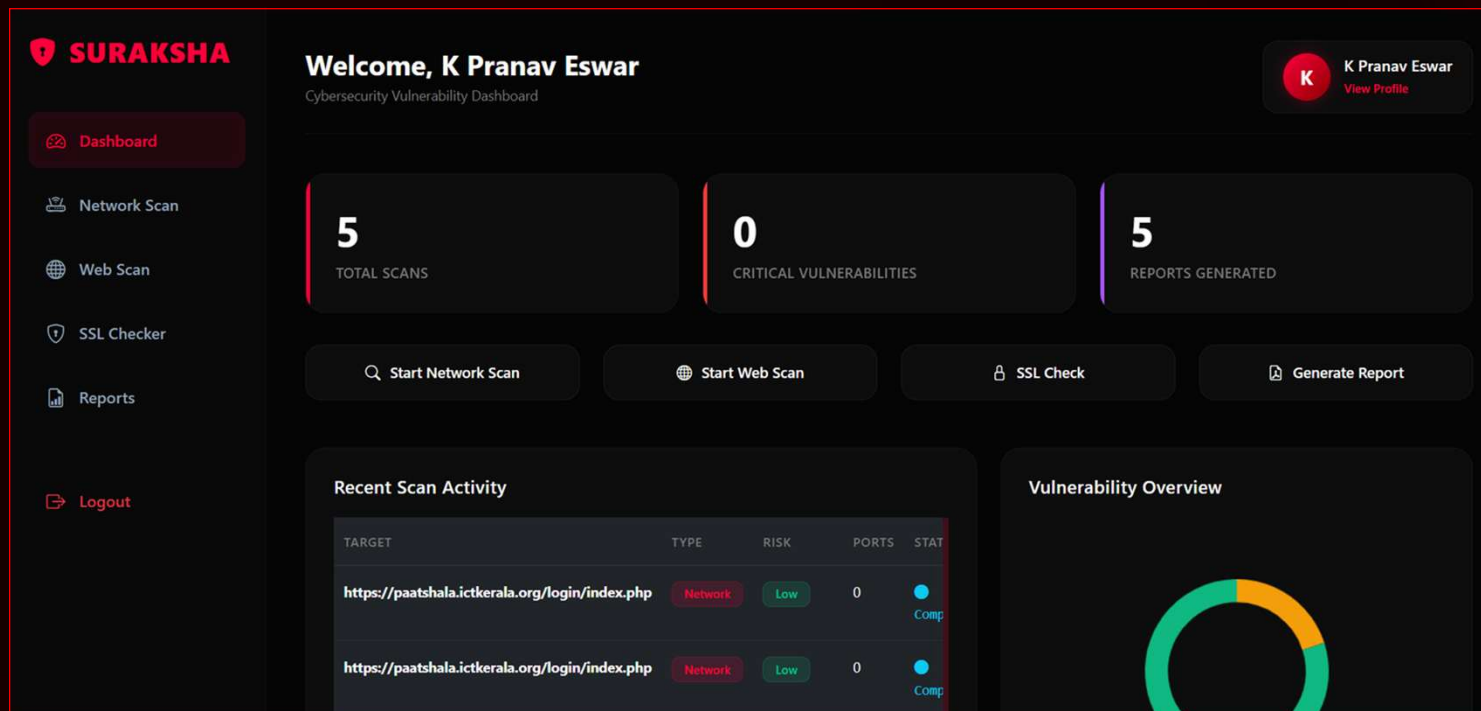


fig. dashboard page

 **SURAKSHA**

 Dashboard

 **Network Scan**

 Web Scan

 SSL Checker

 Reports

 Logout

Network Vulnerability Scanner

ENTER TARGET HOST IP / DOMAIN

scanme.nmap.org

SELECT SCAN TYPE

☐  Port Scan

☐  Banner Grabbing

☐  Service Detection

☐  Version Detection

☐  OS Detection

☐  Vulnerability Scripts

☒  **Full Scan (Select All)**

 Start Network Scan

fig. network scanner page

 **SURAKSHA**

Dashboard

Network Scan

Web Scan

SSL Checker

Reports

Logout

Web Vulnerability Scanner

TARGET APPLICATION URL / WEB ADDRESS

example.com

SELECT WEB SCAN TYPE

☐  Security Headers

☐  Sensitive Paths

☐  SQL Injection

☐  XSS Scan

☐  HTTPS Check

☐  Full Web Scan (Select All)

Start Web Scan

fig. web scanner page

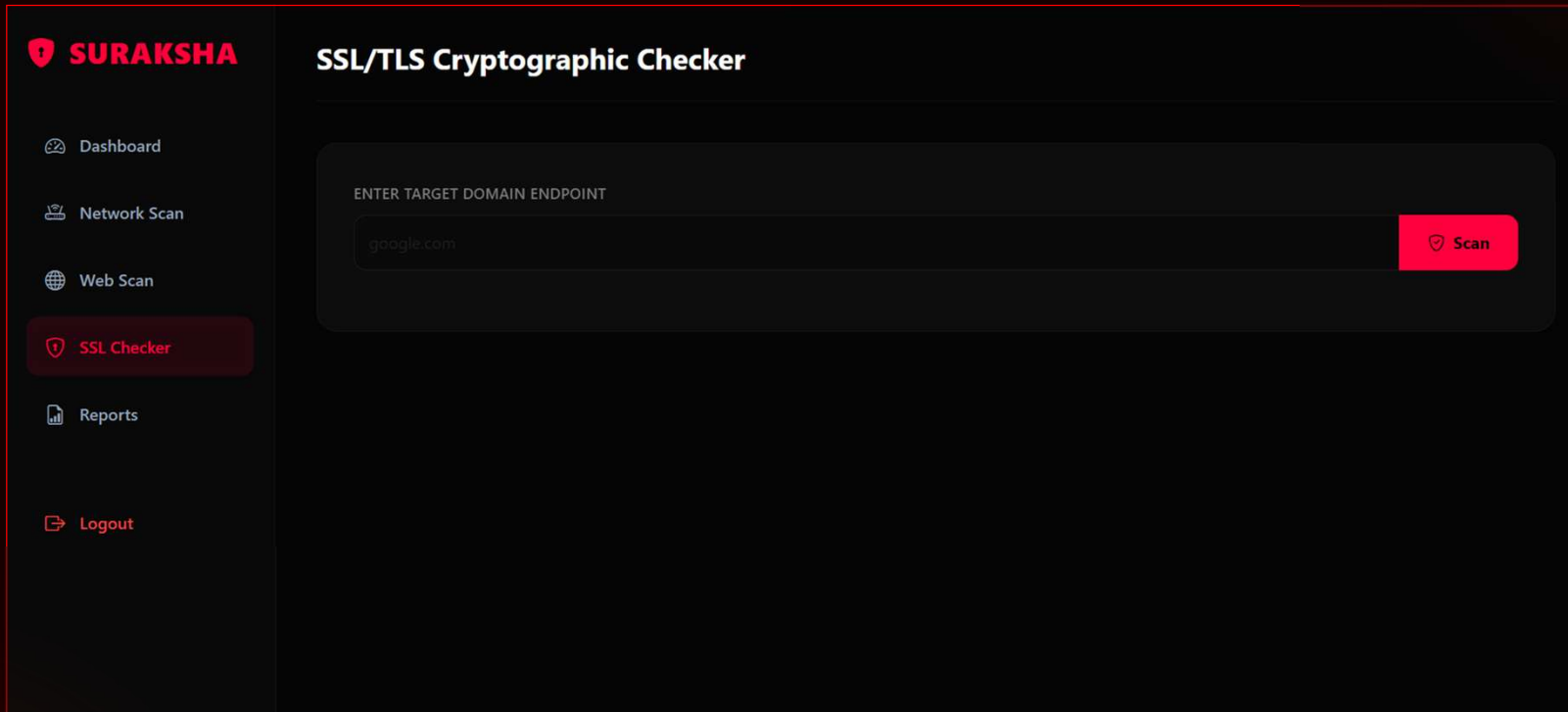


fig. SSL / TLS checker page



Dashboard

Network Scan

Web Scan

SSL Checker

Reports

Logout

All Scan Reports

Search target, domain or IP...

All Scan Types

Search

ID	TARGET SYSTEM	SCAN TYPE	RISK METRIC	DISCOVERED PORTS	EXECUTION TIMESTAMP	ACTIONS
#5	https://paatshala.ictkerala.org/...	Network	Low	0 Active	2026-10-02 07:16:21.530590	View Report
#4	https://paatshala.ictkerala.org/...	Network	Low	0 Active	2026-10-02 06:37:25.252921	View Report
#3	https://paatshala.ictkerala.org/...	Network	Low	0 Active	2026-10-02 06:31:02.803342	View Report
#2	https://paatshala.ictkerala.org/...	Network	Low	0 Active	2026-10-02 06:24:51.605862	View Report

fig. reports page

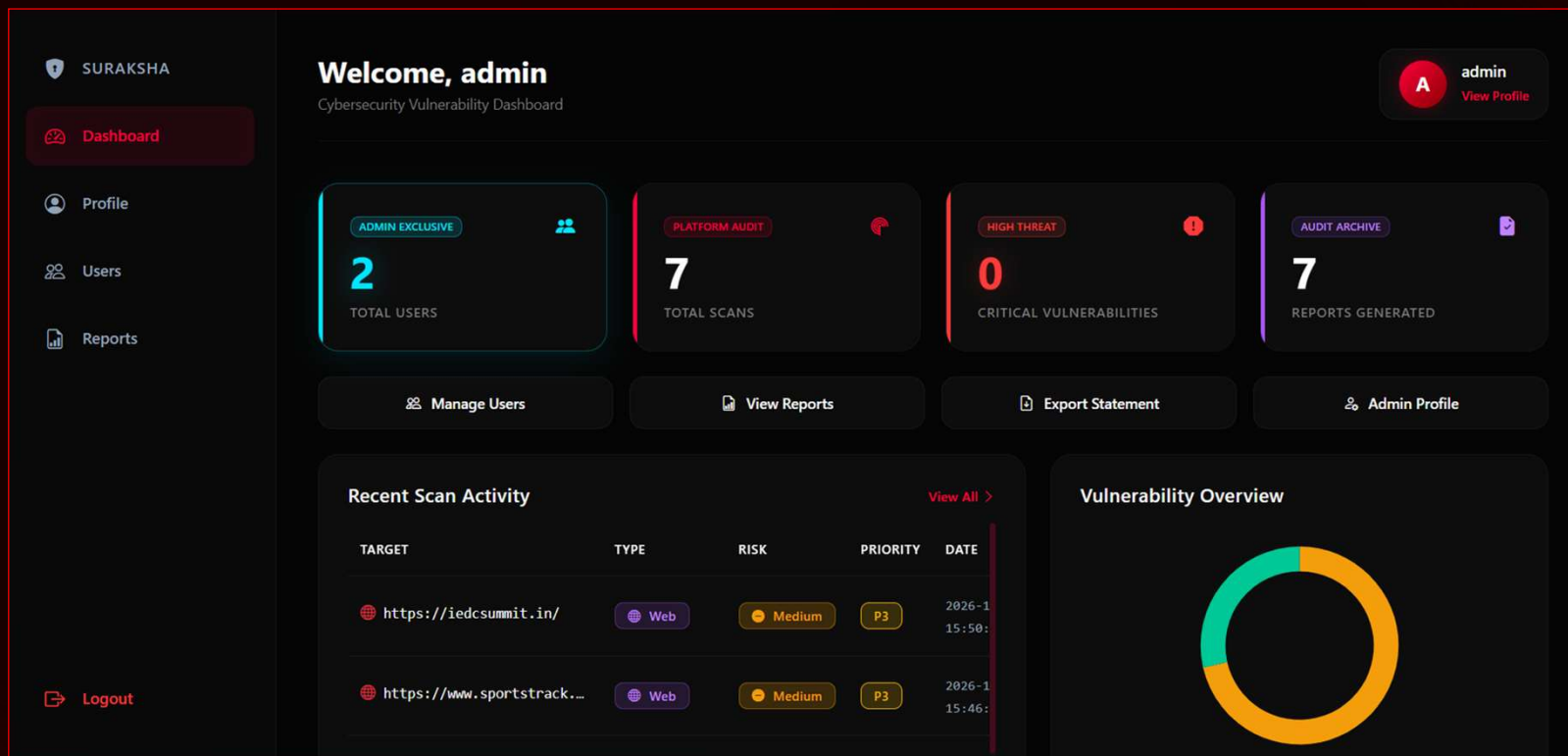


fig. admin dashboard

 SURAKSHA

 Dashboard

 Profile

 Users

 Reports

 Logout

Admin Profile

Administrator Configuration & Profile

Profile Settings

 Edit Credentials

Username

 admin

Email Address

 suraksha@admin.in 

New Password

 Leave blank to keep current password

Cancel

Save Changes

fig. admin
profile

26

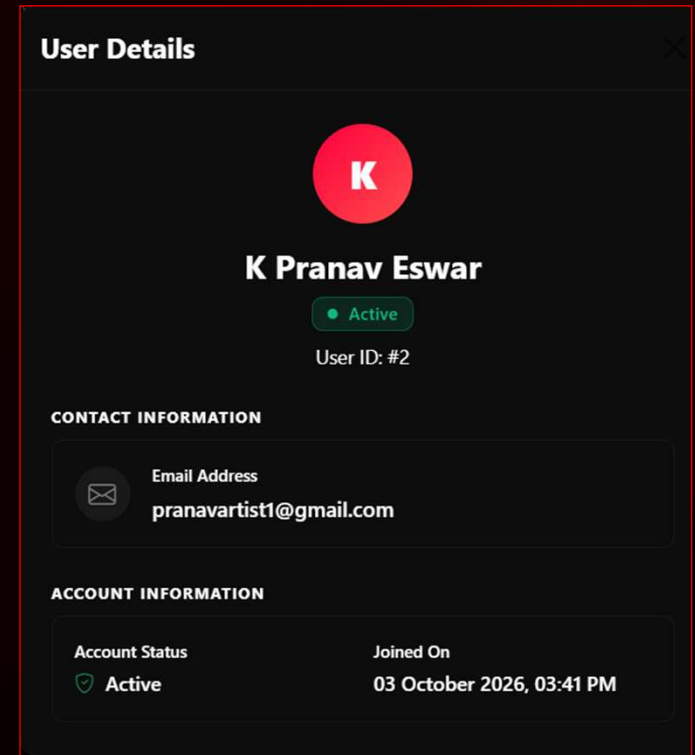
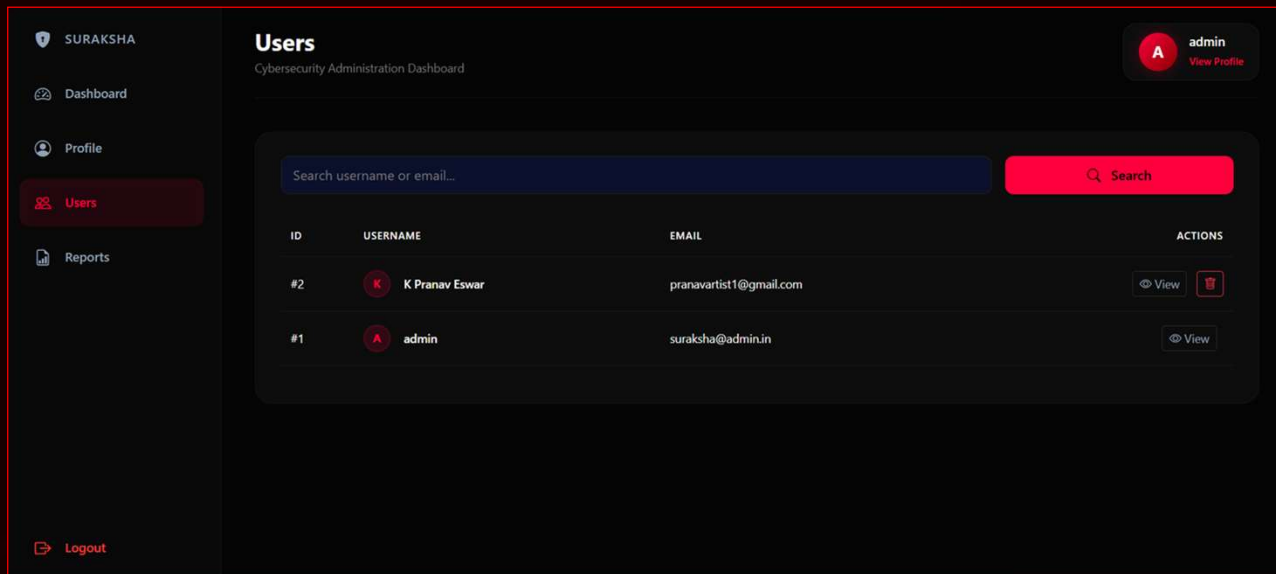


fig. admin / user management

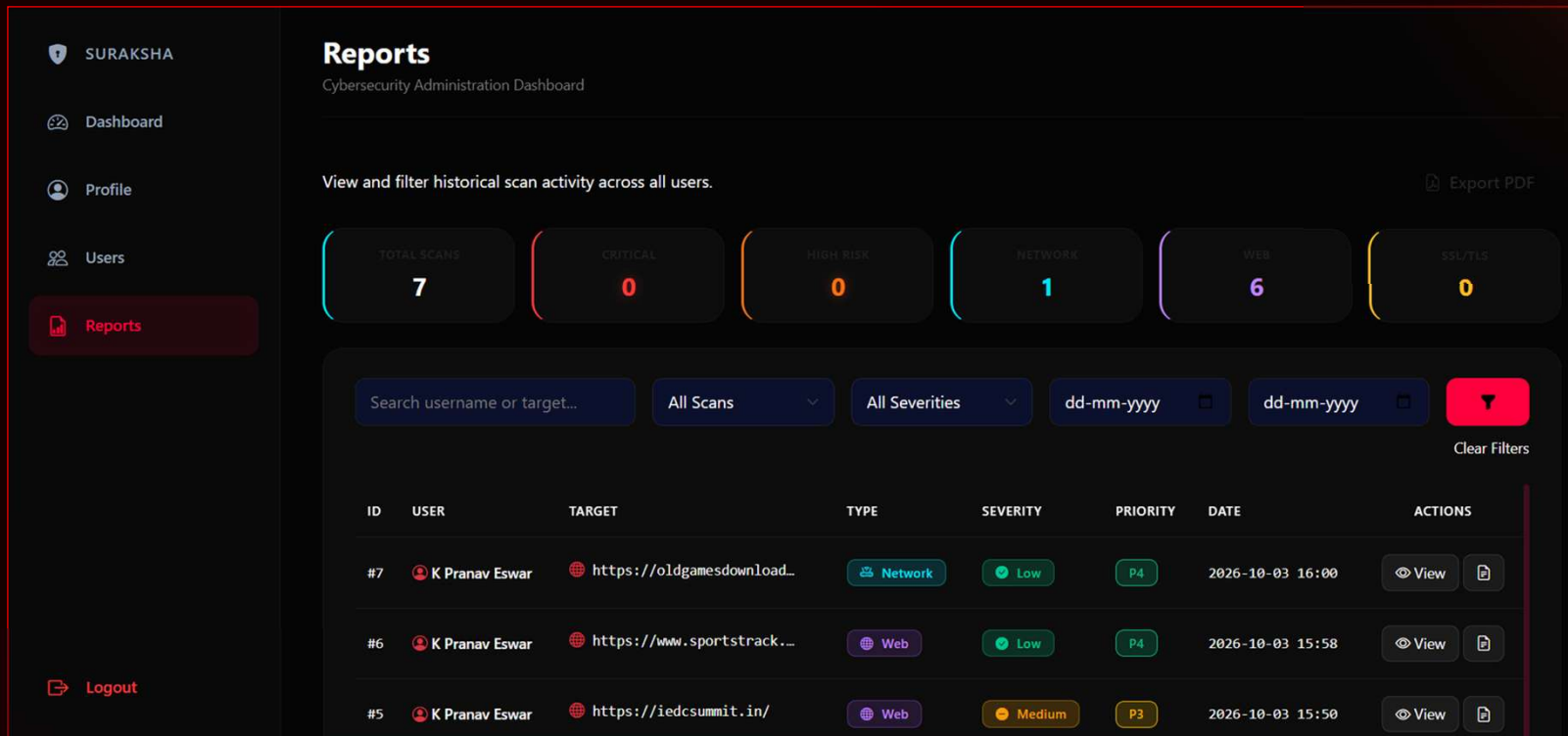


fig. admin reports

RESULTS

Model	Dataset	Accuracy	Precision	Recall	F1-Score
SQL Injection – Random Forest	“sqli.csv” 4,187 samples	96.18%	87.60%	100.00%	93.39%
Network Risk – Random Forest	“network_risk.csv” 3,754 samples	71.77%	72.47%	71.77%	71.83%

- **Integrated Web, Network & SSL/TLS Security Assessment**
- **SQL Injection ML:** 96.18% Accuracy , 93.39% F1-Score
- **Network Risk ML:** 71.77% Accuracy , 71.83% F1-Score
- **Automated CVE Mapping, Risk Classification & Security Reporting**

REFERENCES

- Malkawi, M.; Alhadj, R. *AI-Powered Vulnerability Detection and Patch Management in Cybersecurity: A Systematic Review of Techniques, Challenges, and Emerging Trends*. *Mach. Learn. Knowl. Extr.* 2026, 8, 19. <https://doi.org/10.3390/make8010019>
- Bakır, R. *UniEmbed: A Novel Approach to Detect XSS and SQL Injection Attacks Leveraging Multiple Feature Fusion with Machine Learning Techniques*. *Arab J Sci Eng* 50, 15591–15604 (2025). <https://doi.org/10.1007/s13369-024-09916-4>
- Arasteh, B., Aghaei, B., Farzad, B. et al. *Detecting SQL injection attacks by binary gray wolf optimizer and machine learning algorithms*. *Neural Comput & Applic* 36, 6771–6792 (2024). <https://doi.org/10.1007/s00521-024-09429-z>
- OWASP Foundation. (2021). *OWASP Top 10:2021*. <https://owasp.org/Top10/>
- Lyon, G. (2025). *Nmap Network Scanning: The Official Nmap Project Guide to Network Discovery and Security Scanning*. Nmap Project. <https://nmap.org/book/>

CONCLUSION

- *SURAKSHA provides an integrated platform for web and network vulnerability assessment.*
- *Combines web scanning, network reconnaissance, SSL/TLS analysis, protocol analysis, and CVE mapping.*
- *Uses rule-based security testing and machine-learning-assisted risk prediction.*
- *Provides centralized scan results, severity levels, risk scores, and security reports.*
- *Helps users identify vulnerabilities and understand their potential security impact.*
- *Reduces the need for multiple independent security assessment tools.*

FUTURE ENHANCEMENTS

- *Expand datasets and attack types to improve vulnerability detection.*
- *Detect obfuscated and unseen attacks using improved ML techniques.*
- *Improve CVE updates and security recommendations with regularly updated vulnerability information.*

THANK YOU